

3.5 Options

Unit 5A Astrophysics

In this option, fundamental physical principles are applied to the study and interpretation of the Universe. Candidates will gain deeper insight into the behaviour of objects at great distances from Earth and discover the ways in which information from these objects can be gathered. The underlying physical principles of the optical and other devices used are covered and some indication is given of the new information gained by the use of radio astronomy. Details of particular sources and their mechanisms are not required.

A.1.1 Lenses and Optical Telescopes

- **Lenses**

Principal focus, focal length of converging lens.

Formation of images by a converging lens.

Ray diagrams.

$$\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$$

- **Astronomical telescope consisting of two converging lenses**

Ray diagram to show the image formation in normal adjustment.

Angular magnification in normal adjustment.

$$M = \frac{\text{angle subtended by image at eye}}{\text{angle subtended by object at unaided eye}}$$

Focal lengths of the lenses.

$$M = \frac{f_o}{f_e}$$

- **Reflecting telescopes**

Focal point of concave mirror.

Cassegrain arrangement *using a parabolic concave primary mirror and convex secondary mirror*, ray diagram to show path of rays through the telescope as far as the eyepiece.

Relative merits of reflectors and refractors including a qualitative treatment of spherical and chromatic aberration.

- **Resolving power**

Appreciation of diffraction pattern produced by circular aperture.

Resolving power of telescope, Rayleigh criterion,

$$\theta \approx \frac{\lambda}{D}$$

- **Charge coupled device**

Use of CCD to capture images.

Structure and operation of the charge coupled device:

A CCD is a silicon chip divided into picture elements (pixels).

Incident photons cause electrons to be released.

The number of electrons liberated is proportional to the intensity of the light.

These electrons are trapped in 'potential wells' in the CCD.

An electron pattern is built up which is identical to the image formed on the CCD.

When exposure is complete, the charge is processed to give an image.
Quantum efficiency of pixel > 70%.

A.1.2 Non-optical Telescopes

- **Single dish radio telescopes, I-R, U-V and X-ray telescopes**

Similarities and differences compared to optical telescopes including structure, positioning and use, including comparisons of resolving and collecting powers.

A.1.3 Classification of Stars

- **Classification by luminosity**

Relation between brightness and apparent magnitude.

- **Apparent magnitude, m**

Relation between intensity and apparent magnitude.

Measurement of m from photographic plates and distinction between photographic and visual magnitude not required.

- **Absolute magnitude, M**

Parsec and light year.

Definition of M , relation to m

$$m - M = 5 \log \frac{d}{10}$$

- **Classification by temperature, black body radiation**

Stefan's law and Wien's displacement law.

General shape of black body curves, experimental verification is not required.

Use of Wien's displacement law to estimate black-body temperature of sources

$$\lambda_{\max} T = \text{constant} = 2.9 \times 10^{-3} \text{ mK}.$$

Inverse square law, assumptions in its application.

Use of Stefan's law to estimate area needed for sources to have same power output as the Sun.

$$P = \sigma AT^4$$

Assumption that a star is a black body.

- **Principles of the use of stellar spectral classes**

Description of the main classes:

Spectral Class	Intrinsic Colour	Temperature (K)	Prominent Absorption Lines
O	blue	25 000 – 50 000	He+, He, H
B	blue	11 000 – 25 000	He, H
A	blue-white	7 500 – 11 000	H (strongest) ionised metals
F	White	6 000 – 7 500	ionised metals
G	yellow-white	5 000 – 6 000	ionised & neutral metals
K	orange	3 500 – 5 000	neutral metals
M	red	< 3 500	neutral atoms, TiO

Temperature related to absorption spectra limited to Hydrogen Balmer absorption lines: need for atoms in $n = 2$ state.

- **The Hertzsprung-Russell diagram**

General shape: main sequence, dwarfs and giants.

Axis scales range from -10 to 15 (absolute magnitude) and 50 000 K to 2 500 K (temperature) or OBAFGKM (spectral class).

Stellar evolution: path of a star similar to our Sun on the Hertzsprung-Russell diagram from formation to white dwarf.

- **Supernovae, neutron stars and black holes**

Defining properties: rapid increase in absolute magnitude of supernovae; composition and density of neutron stars; escape velocity $> c$ for black holes.

Use of supernovae as standard candles to determine distances. Controversy concerning accelerating Universe and dark energy.

Supermassive black holes at the centre of galaxies.

Calculation of the radius of the event horizon for a black hole Schwarzschild radius (R_s)

$$R_s = \frac{2GM}{c^2}$$

A.1.4 Cosmology

- **Doppler effect**

$$z = \frac{\Delta f}{f} = \frac{v}{c} \quad \text{and} \quad \frac{\Delta \lambda}{\lambda} = -\frac{v}{c}$$

For $v \ll c$ applied to optical and radio frequencies.

Calculations on binary stars viewed in the plane of orbit, galaxies and quasars.

- **Hubble's law**

Red shift

$$v = Hd$$

Simple interpretation as expansion of universe; estimation of age of universe, assuming H is constant.

Qualitative treatment of Big Bang theory including evidence from cosmological microwave background radiation, and relative abundance of H and He.

- **Quasars**

Quasars as the most distant measurable objects.

Discovery as bright radio sources.

Quasars show large optical red shifts; estimation of distance.

Unit 5B Medical Physics

This option offers an opportunity for students with an interest in biological and medical topics to study some of the applications of physical principles and techniques in medicine.

B.2.1 Physics of the Eye

- **Physics of vision**
Simple structure of the eye.
The eye as an optical refracting system, including ray diagrams of image formation.
- **Sensitivity of the eye**
Spectral response as a photodetector.
- **Spatial resolution**
Explanation in terms of the behaviour of rods and cones.
- **Persistence of vision**
Excluding a physiological explanation.
- **Lenses**
Properties of converging and diverging lenses; principal focus, focal length and power,
$$\text{power} = \frac{1}{f}, \quad \frac{1}{u} + \frac{1}{v} = \frac{1}{f} \quad \text{and} \quad m = \frac{v}{u}$$
- **Ray diagrams**
Image formation.
- **Defects of vision**
Myopia, hypermetropia and astigmatism.
- **Correction of defects of vision using lenses**
Ray diagrams and calculations of powers (in dioptres) of correcting lenses for myopia and hypermetropia.
The format of prescriptions for astigmatism.

B.2.2 Physics of the Ear

- **The ear as a sound detection system**
Simple structure of the ear, transmission processes.
- **Sensitivity and frequency response**
Production and interception of equal loudness curves.
Human perception of relative intensity levels and the need for a logarithmic scale to reflect this.
- **Relative intensity levels of sounds**
Measurement of sound intensity levels and the use of dB and dBA scales.
Definition of intensity.

- **The threshold of hearing**

$$I_0 = 1.0 \times 10^{-12} \text{ Wm}^{-2}$$

$$\text{intensity level} = 10 \log \frac{I}{I_0}$$

- **Defects of hearing**

The effect on equal loudness curves and the changes experienced in terms of hearing loss of:
injury resulting from exposure to excessive noise;
deterioration with age (excluding physiological changes).

B.2.3 Biological Measurement

- **Basic structure of the heart**

The heart as a double pump with identified valves.

- **Electrical signals and their detection; action potentials**

The biological generation and conduction of electrical signals; action potential of a nerve cell; methods of detection of electrical signals at the skin surface.

The response of the heart to the action potential originating at the sino-atrial node; action potential of heart muscle.

- **Simple ECG machines and the normal ECG waveform**

Principles of operation for obtaining the ECG waveform; explanation of the characteristic shape of a normal ECG waveform.

B.2.4 Non-Ionising Imaging

- **Ultrasound imaging**

Reflection and transmission characteristics of sound waves at tissue boundaries, acoustic impedance, attenuation.

Advantages and disadvantages of ultrasound imaging in comparison with alternatives including safety issues and resolution.

Piezoelectric devices

Principles of generation and detection of ultrasound pulses.

A-scan and B-scan

Examples of applications.

- **Fibre optics and Endoscopy**

Properties of fibre optics and applications in medical physics; including total internal reflection at the core-cladding interface; physical principles of the optical system of a flexible endoscope; the use of coherent and non-coherent fibre bundles; examples of use for internal imaging and related advantages.

- **MR Scanner**

Basic principles of MR scanner; cross-section of patient scanned using magnetic fields: hydrogen nuclei excited during the scan emit radio frequency (RF) signals as they de-excite: RF signals detected and processed by a computer to produce a visual image.

Candidates will not be asked about the magnetic fields used in an MR scanner, or about de-excitation relaxation times.

B.2.5 X-ray Imaging

- **X-rays**

The physics of diagnostic X-rays.

- **Physical principles of the production of X-rays**

Rotating-anode X-ray tube; methods of controlling the beam intensity, the photon energy, the image sharpness and contrast and the patient dose.

- **Differential tissue absorption of X-rays**

Excluding details of the absorption processes.

- **Exponential attenuation**

Linear coefficient μ , mass attenuation coefficient μ_m , half-value thickness

$$I = I_0 e^{-\mu x} \quad \mu_m = \frac{\mu}{\rho}$$

- **Image contrast enhancement**

Use of X-ray opaque material as illustrated by the barium meal technique.

- **Radiographic image detection**

Photographic detection with intensifying screen and fluoroscopic image intensification; reasons for using these.

- **CT scanner**

Basic principles of CT scanner; movement of X-ray tube; narrow, monochromatic X-ray beam; array of detectors; computer used to process the signals and produce a visual image. Candidates will not be asked about the construction or operation of the detectors.

Comparisons of ultrasound, CT and MR scans; advantages and disadvantages limited to image resolution, cost and safety issues.

Unit 5C Applied Physics

The option offers opportunities for students to reinforce and extend the work of units PHYA1, PHYA2, PHYA4 and PHYA5 section A of the specification by considering applications in areas of engineering and technology. It embraces rotational dynamics and thermodynamics.

The emphasis should be on an understanding of the concepts and the application of Physics. Questions may be set in novel or unfamiliar contexts, but in all such cases the scene will be set and all relevant information will be given.

C.3.1 Rotational dynamics

- **Concept of moment of inertia**

$$I = \sum mr^2$$

Expressions for moment of inertia will be given where necessary.

- **Rotational kinetic energy**

$$E_k = \frac{1}{2} I \omega^2$$

Factors affecting the energy storage capacity of a flywheel.

Use of flywheels in machines.

- **Angular displacement, velocity and acceleration**

Equations for uniformly accelerated motion:

$$\omega_2 = \omega_1 + \alpha t$$

$$\theta = \omega_1 t + \frac{1}{2} \alpha t^2$$

$$\omega_2^2 = \omega_1^2 + 2\alpha \theta$$

$$\theta = \frac{1}{2} (\omega_1 + \omega_2) t$$

- **Torque and angular acceleration**

$$T = I\alpha$$

- **Angular momentum**

$$\text{angular momentum} = I\omega$$

Conservation of angular momentum.

- **Power**

$$W = T\theta \quad P = T\omega$$

Awareness that, in rotating machinery, frictional couples have to be taken into account.

C.3.2 Thermodynamics and Engines

- **First law of thermodynamics**

$$Q = \Delta U + W,$$

where Q is heat entering the system, ΔU is increase in internal energy and W is work done by the system.

- **Non-flow processes**

Isothermal, adiabatic, constant pressure and constant volume changes

$$pV = nRT$$

adiabatic: $pV^\gamma = \text{constant}$

isothermal: $pV = \text{constant}$

at constant pressure $W = p\Delta V$

Application of first law of thermodynamics to the above processes.

- **The p – V diagram**

Representation of processes on p – V diagram.

Estimation of work done in terms of area below the graph.

Expressions for work done are not required except for the constant pressure case, $W = p\Delta V$

Extension to cyclic processes:

work done per cycle = area of loop.

- **Engine cycles**

Understanding of a four-stroke petrol cycle and a Diesel engine cycle, and of the corresponding indicator diagrams; comparison with the theoretical diagrams for these cycles; a knowledge of engine constructional details is not required; where questions are set on other cycles, they will be interpretative and all essential information will be given; indicator diagrams predicting and measuring power and efficiency

input power = calorific value x fuel flow rate.

Indicated power as

(area of p – V loop) x (no. of cycles/s) x (no. of cylinders).

Output or brake power $P = T\omega$

friction power = indicated power – brake power.

Engine efficiency; overall, thermal and mechanical efficiencies.

Overall efficiency = brake power/input power.

Thermal efficiency = indicated power/input power.

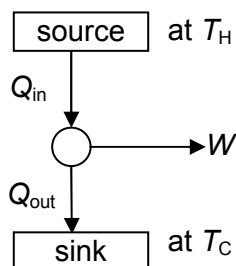
Mechanical efficiency = brake power/indicated power.

- **Second Law and engines**

Need for an engine to operate between a source and a sink

$$\text{efficiency} = \frac{W}{Q_{\text{in}}} = \frac{Q_{\text{in}} - Q_{\text{out}}}{Q_{\text{in}}}$$

$$\text{maximum theoretical efficiency} = \frac{T_{\text{H}} - T_{\text{C}}}{T_{\text{H}}}$$

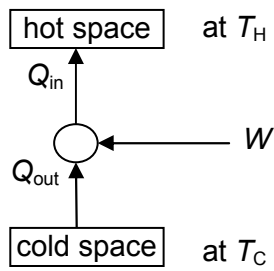


Reasons for the lower efficiencies of practical engines.

Maximising use of W and Q_{out} in combined heat and power schemes.

- **Reversed heat engines**

Basic principles of heat pumps and refrigerators. A knowledge of practical heat pumps or refrigerator cycles and devices is not required.



For a refrigerator: $COP_{ref} = \frac{Q_{out}}{W} = \frac{Q_{out}}{Q_{in} - Q_{out}}$

For a heat pump: $COP_{hp} = \frac{Q_{in}}{W} = \frac{Q_{in}}{Q_{in} - Q_{out}}$

Unit 5D Turning Points in Physics

This option is intended to enable key developments in Physics to be studied in depth so that students can appreciate, from a historical viewpoint, the significance of major conceptual shifts in the subject both in terms of the understanding of the subject and in terms of its experimental basis. Many present day technological industries are the consequence of such key developments and the topics illustrate how unforeseen technologies develop from new discoveries.

D.4.1 The discovery of the Electron

- **Cathode rays**

Production of cathode rays in a discharge tube.

- **Thermionic emission of electrons**

The principle of thermionic emission.

Work done on an electron accelerated through a pd

$$\frac{1}{2}mv^2 = eV$$

- **Determination of the specific charge of an electron, e/m , by any one method**

Significance of Thomson's determination of e/m .

Comparison with the specific charge of the hydrogen ion.

The use of equations

$$F = \frac{eV}{d} \quad F = Bev \quad r = \frac{mv}{Be}$$

- **Principle of Millikan's determination of Q**

Condition for holding a charged oil droplet, of charge Q , stationary between oppositely charged parallel plates

$$\frac{QV}{d} = mg$$

Motion of a falling oil droplet with and without an electric field; terminal speed, Stokes' Law for the viscous force on an oil droplet used to calculate the droplet radius

$$F = 6\pi\eta rv$$

- **Significance of Millikan's results**

Quantisation of electric charge.

D.4.2 Wave Particle Duality

- **Newton's corpuscular theory of light**

Comparison with Huygens' wave theory in general terms.

The reasons why Newton's theory was preferred.

- **Significance of Young's double slits experiment**

Explanation for fringes in general terms, no calculations are expected.

Delayed acceptance of Huygens' wave theory of light.

- **Electromagnetic waves**

Nature of electromagnetic waves.

Maxwell's formula for the speed of electromagnetic waves in a vacuum

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}},$$

where μ_0 is the permeability of free space and ϵ_0 is the permittivity of free space.

Candidates should appreciate that ϵ_0 relates to the electric field strength due to a charged object in free space and μ_0 relates to the magnetic flux density due to a current-carrying wire in free space.

Hertz's discovery of radio waves.

- **The discovery of photoelectricity**

The failure of classical wave theory to explain observations on photoelectricity; the existence of the threshold frequency for the incident light and the variation of the stopping potential with frequency for different metals.

Candidates should appreciate how the stopping potential is measured using a potential divider and a vacuum photocell.

Candidates should also appreciate that photoelectric emission takes place almost instantaneously and that the maximum kinetic energy of the emitted photoelectrons is independent of the intensity of the incident light.

Einstein's explanation of photoelectricity and its significance in terms of the nature of electromagnetic radiation.

- **Wave particle duality**

de Broglie's hypothesis supported by electron diffraction experiments

$$p = \frac{h}{\lambda} \quad \lambda = \frac{h}{\sqrt{2meV}}$$

- **Electron microscopes**

Estimate of anode voltage needed to produce wavelengths of the order of the size of the atom.

Principle of operation of the transmission electron microscope (T.E.M.).

Principle of operation of the scanning tunnelling microscope (S.T.M.).

D.4.3 Special Relativity

- **The Michelson-Morley experiment**

Principle of the Michelson-Morley interferometer.

Outline of the experiment as a means of detecting absolute motion.

Significance of the failure to detect absolute motion.

The invariance of the speed of light.

- **Einstein's theory of special relativity**

The concept of an inertial frame of reference.

The two postulates of Einstein's theory of special relativity:

- (i) physical laws have the same form in all inertial frames,
- (ii) the speed of light in free space is invariant.

- **Time dilation**

Proper time and time dilation as a consequence of special relativity.

Time dilation

$$t = t_0 \left(1 - \frac{v^2}{c^2} \right)^{-\frac{1}{2}}$$

Evidence for time dilation from muon decay.

- **Length contraction**

Length of an object having a speed v

$$l = l_0 \left(1 - \frac{v^2}{c^2} \right)^{\frac{1}{2}}$$

- **Mass and energy**

Equivalence of mass and energy

$$E = mc^2 \qquad E = \frac{m_0 c^2}{\left(1 - \frac{v^2}{c^2} \right)^{\frac{1}{2}}}$$